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**FAST National University of Computer
and Emerging Sciences**

Karachi Campus

PROJECT PROPOSAL

Console Quiz

An object-oriented console application for managing classrooms, quizzes, timed attempts, and leaderboards

Course : Object Oriented Programming (OOP)
Instructor : Sir Muhammad Khalid Khan
Section : BAI-2B
Submission Date : April 2026

Group Members

Member 1 : Ahmed Asif 25K-0039
Member 2 : Muhammad Ahmed 25K-0105
Member 3 : Ahmad Ali Tariq 25K-0138

1. Project Title

Console Quiz (CQ) — A C++ object-oriented application for managing virtual classrooms, interactive quizzes with multiple choice questions, timed attempts, student scoring, and automated leaderboard generation using core OOP principles.

2. Group Members

#	Name	Roll Number	Responsibility
1	Ahmad Ali Tariq	25K-0138	Class Design, Inheritance & Polymorphism
2	Ahmed Asif	25K-0039	Quiz Logic, Memory Management & Testing
3	Muhammad Ahmed	25K-0105	UI/Menu System, Leaderboard & Documentation

3. Problem Statement

Educational institutions currently rely on manual methods or basic digital platforms that do not provide specialized classroom management and automated timed assessments. This results in difficulty for instructors to manage student enrollment for specific quizzes and manual overhead in calculating marks and ranking performance.

An object-oriented software solution is required to model entities like Teachers, Students, Classrooms, and Quizzes as classes with clear boundaries. The system must enforce business rules like secure classroom access, timed attempt constraints, and provide persistent-like logic for scoring students within a classroom session.

4. Proposed Solution

We propose to build a Quiz Application in C++ that utilizes all core OOP pillars. The system features a menu-driven console interface with distinct functional modules for Teachers and Students.

Key Modules & Classes:

- **User (Abstract Base):** Defines common properties for all users (name, ID, password) and enforces a uniform setup via pure virtual methods.
- **Teacher & Student (Derived):** Specialize user behavior; Teachers manage classes and quizzes, while Students join classes and take assessments.
- **Classroom:** Orchestrates the interaction between a teacher and their enrolled students, storing specific quizzes and tracking performance.
- **Quiz & Question:** Handles the structure of assessments, including multiple-choice options and correct answer matching logic.
- **Attempt:** A session-manager class that utilizes system time to enforce quiz durations and calculate student percentages.
- **System:** The top-level manager that handles dynamic memory for all objects and facilitates the main application flow.

5. OOP Concepts to be Implemented

OOP Concept	Where Applied
Classes & Objects	Teacher, Student, Classroom, Quiz, Question, Attempt, System
Encapsulation	Private data members and logic validation (e.g., Question options)
Inheritance	Student and Teacher classes inherit from the User base class

OOP Concept	Where Applied
Polymorphism	Virtual destructor and pure virtual setUser() method in User class
Abstract Classes	User class serves as an abstract base for user identity
Friend Classes	Granting System access to private fields of Teacher/Student; Attempt to Quiz
STL Containers	Extensive use of std::vector for managing dynamic collections of objects
Memory Management	Manual management using new/delete within the System class
Static Members	Static ID counters (idn, idnum) to ensure unique IDs for all users
Time Handling	Using <ctime> library for real-time quiz timing and time-up checks

6. Technology Stack

Tool / Technology	Purpose
C++ (C++17 standard)	Primary programming language
VS Code / Dev-C++	Development Environment
STL (Standard Template Library)	Vector and String handling
Ctime Library	Implementation of quiz timers
Git / GitHub	Version control

7. Project Timeline

Week	Milestone	Deliverable
Week 1	Proposal & Class Design	Project proposal document & Initial Class Structure
Week 2	Base Implementation	User, Teacher, and Student class hierarchies
Week 3	Quiz Logic	Question structures and validation logic
Week 4	Classroom Module	Enrollment and class-specific data management
Week 5	Attempt Session	Timer implementation and scoring algorithms
Week 6	Menu System	Integration of all modules into a menu-driven interface
Week 7	Testing & Submission	Final report, code cleanup, and viva preparation

8. Marking Scheme (Reference)

Component	Marks	Deadline
Proposal	1	Week 1
Code Submission	3	Before viva
Project Report	2	Before viva
Viva Voce	4	28 April 2026
Total	10	—

9. Division of Work

Member	Roll No.	Tasks Assigned
Ahmad Ali Tariq	25K-0138	Class hierarchy, inheritance structure, polymorphism, and UML design.
Ahmed Asif	25K-0039	Quiz logic, Attempt session timer, and scoring algorithms.
Muhammad Ahmed	25K-0105	Menu-driven UI, System integration, and documentation.

10. Expected Outcome

Upon completion, the Quiz Application will provide:

- A fully functional C++ console application demonstrating all core pillars of Object-Oriented Programming.
- A secure classroom environment where teachers can manage assessments and students can participate via secure codes.
- Automated quiz timers ensuring fair assessment conditions for all students.
- Live leaderboard generation to track student performance and rankings.
- Modular code structure following professional standards with header and source file separation.
- Comprehensive memory management ensuring zero leaks using proper destructors and manual deletion.

Note: All group members are aware of and understand every component of the project. AI tools may be used for coding assistance, but each member must be able to explain, modify, and extend their assigned code during the viva evaluation.

Student Representative Signature

Ahmed Asif | 25K-0039 (LEAD)

Instructor Approval

Sir Muhammad Khalid Khan